

MODULE – V COMPLETE ANSWERS (DBMS)

Based on the uploaded question bank.

SHORT ANSWER QUESTIONS (5 MARKS)

1. Explain the Pros and Cons of Sequential File Organization

Sequential File Organization

Records are stored one after another in a specific order, usually based on a key field.

Diagram

```
+-----+ +-----+ +-----+ +-----+  
| R1   | → | R2   | → | R3   | → | R4   |  
+-----+ +-----+ +-----+ +-----+
```

Advantages

1. Simple Structure

Easy to implement and maintain.

2. Efficient for Sequential Access

Useful when processing all records.

3. Less Storage Overhead

No extra index required.

4. Suitable for Batch Processing

Payroll systems, report generation.

Disadvantages

1. Slow Random Access

Must scan records sequentially.

2. Expensive Insertions

New records may require reorganization.

3. Expensive Deletions

File restructuring may be needed.

4. Poor Performance for Frequent Updates

Not suitable for dynamic databases.

2. Difference Between B-Tree and B+ Tree

Feature	B-Tree	B+ Tree
Data Storage	Internal and Leaf Nodes	Only Leaf Nodes
Search Speed	Lower	Faster
Sequential Access	Difficult	Easy
Leaf Node Link	Not Present	Present
Fan-Out	Lower	Higher
Disk Access	More	Less

Range Queries

Less Efficient

More Efficient

B-Tree Example

```
  20
 /  \
10   30
```

Data can exist in all nodes.

B+ Tree Example

```
  20
 /  \
10   30
```

Leaf:

10 → 20 → 30

Only leaf nodes store actual records.

3. Explain Hash File Organization with Suitable Diagram

Hashing stores records using a hash function.

Formula

Address = $h(\text{Key})$

The hash function determines the storage location.

Example

Suppose:

$h(\text{Key}) = \text{Key} \bmod 10$

For:

Key = 123

$123 \bmod 10 = 3$

Store record in bucket 3.

Diagram

Key
|
Hash Function
|
Bucket Number
|
Storage Location

Advantages

- Fast search
 - Fast insertion
 - Constant average access time
-

Disadvantages

- Collision problem
 - Poor range query performance
-

4. Define Seek Time, Access Time and Rotational Latency

Seek Time

Time required to move disk arm to the desired track.

Formula

Seek Time = Time to reach track

Rotational Latency

Time waiting for required sector to rotate under read/write head.

Formula

Average Rotational Latency
= $\frac{1}{2}$ Rotation Time

Access Time

Total time needed to access data.

Formula

Access Time
=
Seek Time
+
Rotational Latency
+
Transfer Time

Example

Seek Time = 8 ms
Latency = 4 ms
Transfer = 2 ms

Access Time = 14 ms

5. Compare Primary Indexing and Clustering Indexing

Feature	Primary Index	Clustering Index
Ordering Field	Primary Key	Non-Key Field
Uniqueness	Unique	Not Necessary
Records Sorted By	Primary Key	Clustering Field
Number Allowed	One	One
Search Efficiency	Very High	High

Example

Primary Index

Student(SID, Name)

Records sorted by SID.

Clustering Index

Employee(DeptNo)

Records sorted by DeptNo.

Multiple employees may have same DeptNo.

6. Disadvantages of B-Tree over B+ Tree

1. Lower Fan-Out

Stores data in internal nodes.

2. Larger Tree Height

More levels may be needed.

3. More Disk Access

Searches require more node accesses.

4. Poor Sequential Traversal

Leaf nodes are not linked.

5. Less Efficient Range Queries

Compared to B+ Tree.

7. Explain How Secondary Indexing Helps in Join Operations

Secondary Index

An index built on a non-ordering attribute.

Example

Employee

EmpID	DeptID
1	10
2	20

Department

DeptID	Name
10	CSE
20	EE

Join:

```
SELECT *  
FROM Employee E  
JOIN Department D  
ON E.DeptID=D.DeptID;
```

If a secondary index exists on DeptID:

Index(DeptID)

DBMS quickly finds matching rows.

Advantages

- Faster joins
 - Reduced scanning
 - Better query performance
-

LONG ANSWER QUESTIONS (10 MARKS)

1. Explain Sequential File Organization with Suitable Diagram

Sequential organization stores records in sorted order.

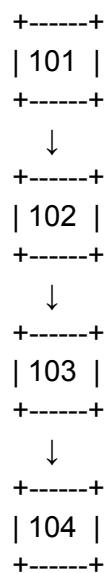
Example

Student Records

101
102
103
104
105

Stored sequentially.

Diagram



Operations

Search

Sequential scan.

Insert

Locate proper position and insert.

Delete

Locate and remove record.

Advantages

1. Easy implementation
 2. Suitable for sequential processing
 3. Efficient batch operations
 4. Low storage overhead
-

Disadvantages

1. Slow random access
 2. Expensive insertion
 3. Expensive deletion
 4. File reorganization required
-

Applications

- Payroll Systems
 - Banking Reports
 - Student Result Processing
-

2. Construct B+ Tree

Insertion Sequence:

2, 3, 5, 11, 17, 19, 23, 29, 31

Maximum pointers per node:

$m = 4$

Step 1

Insert

2 3 5

Leaf:

[2 3 5]

Step 2

Insert 11

Overflow occurs.

Split.

```
    [5]
   /  \
 [2 3] [5 11]
```

Step 3

Insert 17

```
    [5]
   /  \
 [2 3] [5 11 17]
```

Step 4

Insert 19

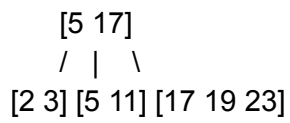
Overflow.

Split.

```
    [5 17]
   /  |  \
 [2 3] [5 11] [17 19]
```

Step 5

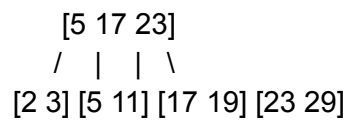
Insert 23



Step 6

Insert 29

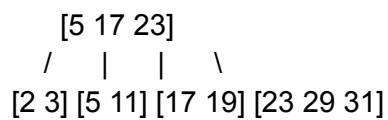
Split leaf.



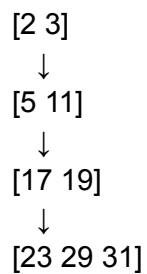
Step 7

Insert 31

Final B+ Tree



Leaf Links



3. Number of Blocks for Data File and Dense Index

Given:

One block stores 3 records

One index block stores 10 key pointers

Database contains n records

Data Blocks

Each block stores 3 records.

Therefore:

$$\text{Data Blocks} = \left\lceil \frac{n}{3} \right\rceil$$

Dense Index Blocks

Dense index contains one entry per record.

Number of entries:

n

Each index block stores 10 entries.

Therefore:

$$\text{Index Blocks} = \left\lceil \frac{n}{10} \right\rceil$$

Final Answer

Data Blocks = $\lceil n/3 \rceil$

Dense Index Blocks = $\lceil n/10 \rceil$

4. Order of Leaf Node in B+ Tree

Given:

Block Size = 1 KB
= 1024 Bytes

Record Pointer = 7 Bytes

Value Field = 9 Bytes

Block Pointer = 6 Bytes

Formula

For leaf node:

$$\begin{aligned} &m(\text{value} + \text{pointer}) \\ &+ \\ &\text{block pointer} \\ &\leq \text{block size} \end{aligned}$$

Substitute values:

$$m(9 + 7) + 6 \leq 1024$$

$$16m + 6 \leq 1024$$

$$16m \leq 1018$$

$$m \leq 63.625$$

Therefore:

$$m = 63$$

Answer

Order of Leaf Node = 63

EXAM REVISION (VERY IMPORTANT)

Frequently Asked Questions

1. Sequential File Organization
2. B-Tree vs B+ Tree
3. Hashing
4. Primary Index vs Clustering Index
5. Dense and Sparse Index
6. Seek Time and Rotational Latency
7. B+ Tree Construction
8. Order of B+ Tree Node
9. Secondary Indexing
10. File Organization Techniques

Most Important Numericals

- B+ Tree Construction
- Dense Index Block Calculation
- Order of Leaf Node
- Hash Function Problems

These are the highest-scoring topics from Module V.